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2. Introduction

2.1 Brief Overview of the Network Scenario

This project focuses on upgrading a small-to-medium enterprise from an inefficient, flat network topology to a structured and secure hierarchical architecture. In this current setup, all host devices, departmental workstations, and management endpoints across the entire organization are placed within a single, unsegmented broadcast domain. This creates a highly inefficient network environment where excessive broadcast traffic flows to every connected device, degrading network performance and wasting valuable bandwidth. More critically, there are no logical controls to isolate sensitive administrative systems from student areas or unauthenticated public guest devices. The legacy network also lacks automated IP management, link redundancy, and access-layer protection, leaving the organization highly vulnerable to operational downtime, human configuration error, and unauthorized network intrusions.

2.2 Objectives of the Network Design

The main goal of our network design is focused on building a secure and reliable network that fully meets all the requirements outlined.

First, we will completely eliminate the insecure flat network layout by creating four separate department VLANs (VLAN 10 Staff/Admin, VLAN 20 Student/User, VLAN 30 Guest, and VLAN 99 Management) to isolate traffic, while setting up dynamic DHCP pools on our core router to automate IP configurations.

Second, we plan to remove single points of failure by bundling the physical links between our switches into a high-bandwidth LACP EtherChannel, while configuring PVST+ to maintain a completely loop-free topology with manual Root Bridge placement.

Third, we aim to heavily lock down our access layer by enabling sticky port security, disabling DTP to stop trunk-spoofing, and moving all unused switch ports into a dead-end VLAN before shutting them down completely.

Fourth, we will protect our actual network hardware by changing the native VLAN away from VLAN 1, creating local admin accounts, and forcing encrypted SSH access instead of insecure Telnet for remote management.

Finally, we will establish working internet connectivity by configuring a static default route to our simulated ISP router along with a return route back to our internal subnets.

3. Network Requirements

3.1 Technical Requirement List

Physically, our topology must include at least two Cisco switches to handle local connections and at least two Cisco routers, one acting as our internal gateway and another simulating an external ISP network. Logically, we need to create four specific VLANs to separate departmental traffic: Staff (VLAN 10), Student/User (VLAN 20), Guest/Public (VLAN 30) , and a secure Management network (VLAN 99). To automate how devices get their network settings, our internal core router needs to run centralized DHCP pools for each VLAN, while keeping gateway and switch management IPs safely excluded from being handed out.

We also need to configure 802.1Q trunking on all links carrying traffic for multiple VLANs, which includes the connection between the two switches and the uplink to the router, while ensuring we change the native VLAN away from the default setting. To keep the network stable and fast, we must bundle the inter-switch links into an LACP EtherChannel for extra bandwidth and backup paths, using Spanning Tree Protocol (STP) to prevent dangerous layer 2 loops. For access-layer security, we are required to implement at least four security features. We will achieve this by enabling sticky port security on all access ports, administratively shutting down every unused port after assigning it to an unused dead-end VLAN, statically disabling DTP negotiation on all switch ports, and moving the native VLAN away from VLAN 1, and we further reinforce the access layer with PortFast and BPDU Guard, giving us more than the four required features. Finally, we have to protect our hardware by setting up custom hostnames, secret passwords, local user accounts, and forcing encrypted SSH access, while using a static default route and a matching return route to establish proper internet connectivity with the ISP router.

3.2 Explanation of Selected SRWE Concepts

3.2.1 Virtual Local Area Networks (VLANS) and Trunking

VLANs divide a single physical switch into separate logical networks to shrink broadcast domains, improving network speeds and keeping sensitive department traffic isolated. To carry this separated traffic across multiple switches, 802.1Q Trunking is configured on the connecting links to tag each data frame with its specific VLAN ID. As a security measure, the Native VLAN, which carries untagged frames across a trunk, is moved away from the default VLAN 1 to prevent common hacking techniques like VLAN hopping attacks. (Cisco Networking Academy, 2020, Module 3)

3.2.2 Router on a Stick (ROAS) Inter-VLAN Routing

Since VLANs would completely isolate subnets, a router is required to let different departments communicate securely. Router on a Stick achieves this efficiently by using just one physical interface on the core router, dividing it into multiple virtual subinterfaces (like G0/0.10 and G0/0.20). Each subinterface is configured with 802.1Q encapsulation to read and route traffic for its specific VLAN, serving as that subnet's default gateway. (Cisco Networking Academy, 2020, Module 4)

3.2.3 Dynamic Host Configuration Protocol (DHCP)

Manually assigning static IP addresses to every host is highly inefficient and often causes duplicate IP conflicts. To automate this process, the core router is configured as a centralized DHCP server hosting

dedicated address pools for each departmental VLAN. When an end device plugs in, it automatically receives its IP address, subnet mask, default gateway, and DNS settings. Critical infrastructure IPs are explicitly excluded from these pools to prevent them from being accidentally handed out to user devices. (Cisco Networking Academy, 2020, Module 7)

3.2.4 LACP EtherChannel and Spanning Tree Protocol (STP)

To ensure high availability, redundant physical paths are built between switches, though these extra links can cause network-crashing Layer 2 loops. Spanning Tree Protocol (STP) prevents this by tracking the topology, electing a Root Bridge, and blocking redundant links to keep the environment loop-free. We combine this with LACP EtherChannel, which bundles the physical connections into a single high-speed logical link to multiply bandwidth. Because STP treats the bundle as one logical link, none of the member ports are blocked, and if one physical cable fails, traffic instantly redistributes across the remaining links without waiting for STP reconvergence. (Cisco Networking Academy, 2020, Modules 5-6)

3.2.5 Secure Device Management

Network devices themselves must be hardened against unauthorized access. Each device is given a descriptive hostname and an encrypted enable secret, while the console line is protected by a line password and the VTY lines are locked behind local username authentication. Remote management is restricted to SSH version 2, which requires a configured domain name and an RSA key pair, so every administrative session is encrypted instead of travelling in plaintext the way Telnet does. Each switch is also assigned a management IP address on a Switched Virtual Interface (SVI) inside the dedicated Management VLAN 99, together with a default gateway, so that administrators can reach the switches remotely from the Admin network while ordinary users on other VLANs have no direct path to the management plane. It should be noted that the older Catalyst 2960 switches available in the physical lab run an IOS image without the cryptographic feature set and therefore cannot support SSH; on that equipment the same management plane is protected by console and VTY line passwords, an enable secret, local user accounts and the isolated Management VLAN, while full SSH version 2 operation is demonstrated in the Packet Tracer implementation (see Section 8).

4. Network Topology

4.1 Packet Tracer Topology Diagram

The proposed topology follows a simple two-tier design: two Catalyst 2960 access switches provide user connectivity, and a single ISR 4331 core router (R1) performs all inter-VLAN routing using Router-on-a-Stick, acts as the centralized DHCP server, and forwards internet traffic to a second ISR 4331 that simulates the ISP. A server behind the ISP router simulates a public web and DNS service so that internet connectivity, default routing, and the return route can all be verified with realistic end-to-end tests.

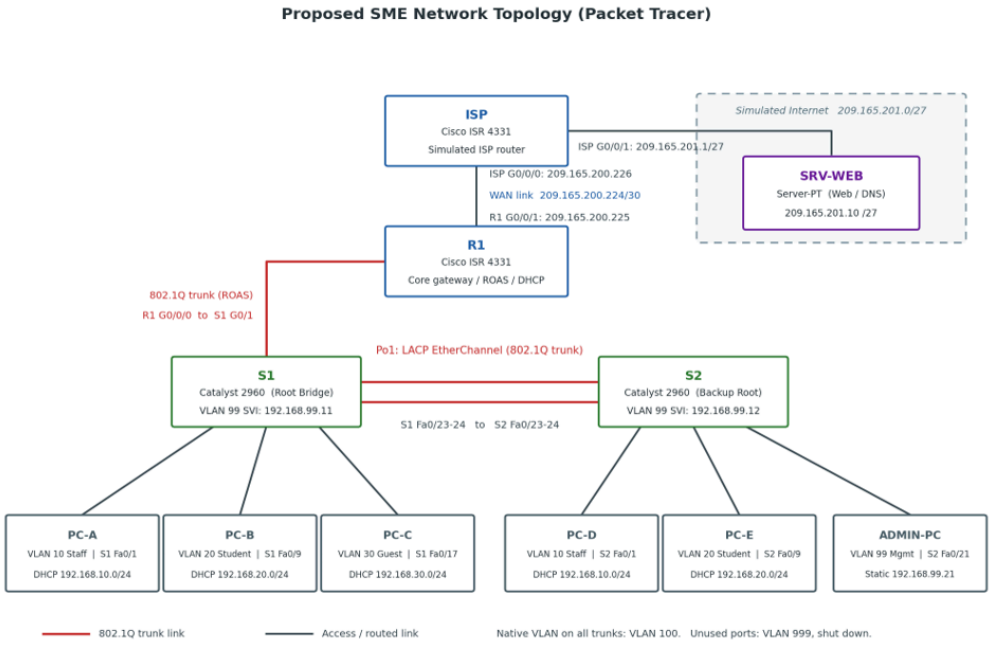


Figure 4.1: Proposed network topology for the Packet Tracer implementation.

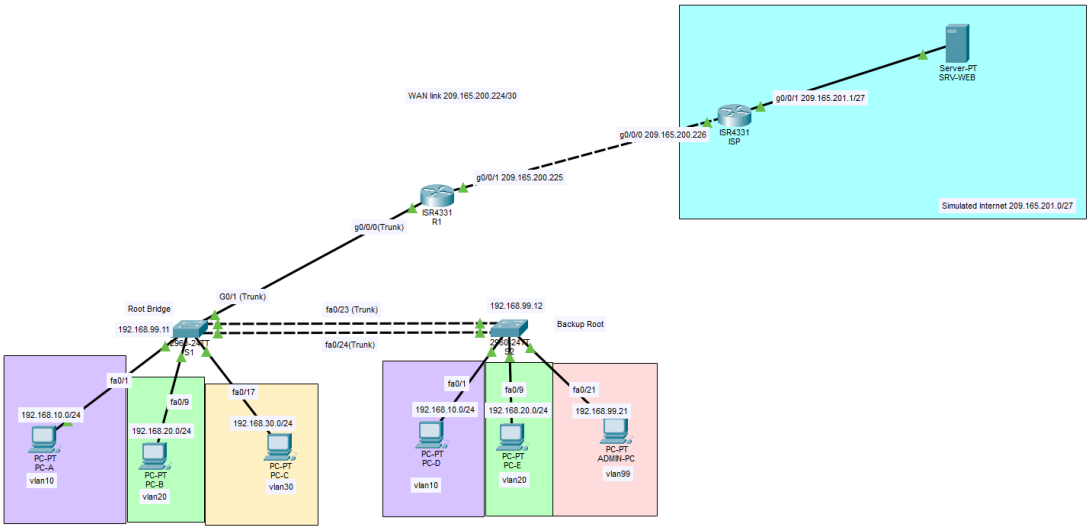


Figure 4.2: Logical network topology for the Packet Tracer implementation.

4.2 Physical Lab Topology

The same logical design will be replicated on the physical Cisco equipment in the lab during the weeks commencing 20 July and 27 July. The cabling plan mirrors Figure 4.1, and the **differences** from the physical lab setup to the packet tracer setup are the **older version** of Cisco Catalyst 2960 Series, which doesn't support SSH, and **Cisco 2911 routers**. There's is also an **absence of a web server** in the physical lab, instead we use the ISP router to do loopback for routing verification.

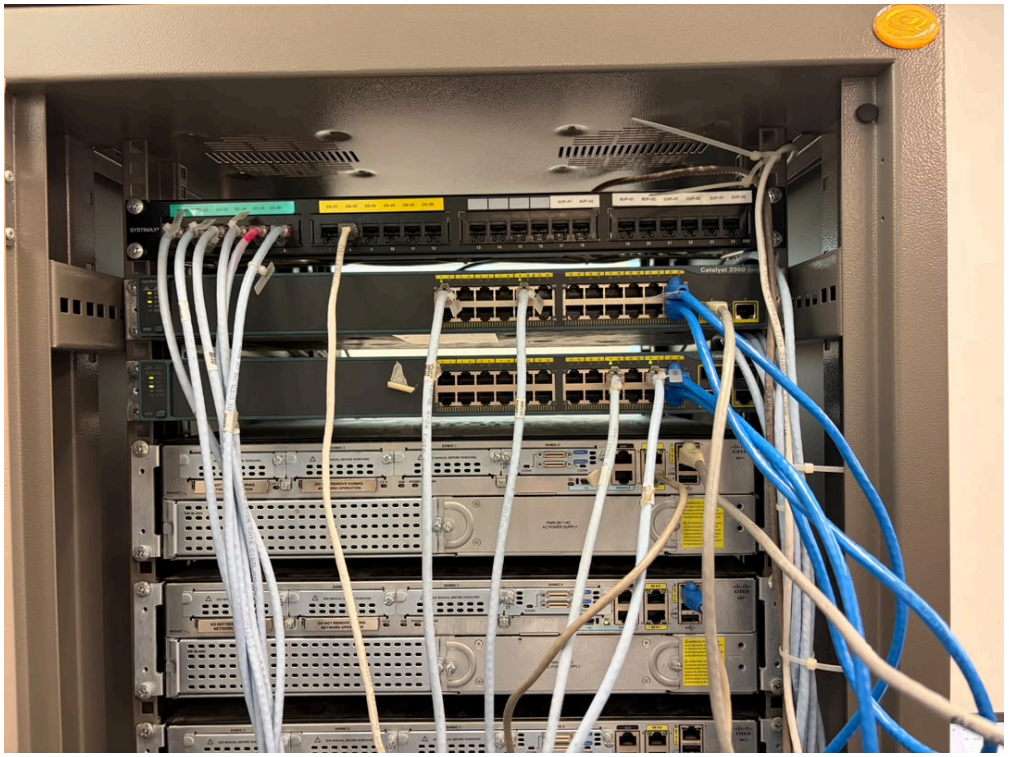


Figure 4.3: Physical network topology for the Lab implementation.

4.3 Device Roles and Selection Justification

Device	Model	Role in the Network
R1	Cisco ISR 4331 (Packet Tracer); Cisco 2911 in physical lab	Core gateway. Hosts the 802.1Q subinterfaces for Router-on-a-Stick inter-VLAN routing, runs the centralized DHCP pools, and holds the default route toward the ISP.
ISP	Cisco ISR 4331 (Packet Tracer); Cisco 2911 in physical lab	Simulates the external ISP. Terminates the 209.165.200.224/30 WAN link and holds the static return route back to the internal 192.168.0.0/16 networks.

S1	Catalyst 2960 (24-port)	Access switch and primary STP Root Bridge for all VLANs. Carries the 802.1Q trunk to R1 and one side of the LACP EtherChannel.
S2	Catalyst 2960 (24-port)	Access switch and secondary (backup) Root Bridge. Hosts the Management VLAN admin workstation.
PC-A to PC-E	Generic PC	DHCP client test hosts. VLAN 10 and VLAN 20 each have one host on each switch to prove that VLANs span both switches; VLAN 30 has one guest host.
ADMIN-PC	Generic PC	Statically addressed management workstation in VLAN 99, used to verify SSH access to all devices in Packet Tracer, and console/password-based management access in the physical lab.
SRV-WEB	Server-PT	Simulated public web and DNS server (209.165.201.10) used as the target for internet connectivity and DNS tests.

Device selection and physical lab adaptations: the Cisco ISR 4331 and the Catalyst 2960 were selected in Packet Tracer because they are the most suitable platforms for this design: the ISR 4331 supports 802.1Q subinterfaces, centralised DHCP and the cryptographic feature set required for SSH version 2, while the 2960 provides the access-layer VLAN, EtherChannel, PVST+ and port-security features within a realistic small-enterprise budget. The equipment available in the teaching laboratory imposed three documented substitutions, none of which changes the logical design. First, Cisco 2911 routers were used in place of the ISR 4331; the only practical difference is the interface naming, so the Router-on-a-Stick parent interface becomes GigabitEthernet0/0 with subinterfaces G0/0.10, G0/0.20, G0/0.30 and G0/0.99. Second, the laboratory Catalyst 2960 switches run an older IOS image without the cryptographic feature set and therefore cannot support SSH, as recorded in Section 8. Third, only four PCs were available, so the end devices were allocated to preserve full test coverage with one host in every VLAN: PC-1 on S1 Fa0/1 (VLAN 10), PC-2 on S1 Fa0/9 (VLAN 20), PC-3 on S2 Fa0/17 (VLAN 30) and the statically addressed ADMIN-PC on S2 Fa0/21 (VLAN 99). No switch or router configuration required alteration for the reduced device count, because each departmental port range was already provisioned on both switches; only the number of cabled ports differed. The EtherChannel remained fully exercised in the laboratory because all traffic from the S2 hosts must traverse Port-channel 1 to reach any gateway.

5. VLAN and IP Addressing Plan

5.1 VLAN Table

VLAN ID	Name	Purpose	Subnet	Default Gateway
10	STAFF	Staff department	192.168.10.0/24	192.168.10.1
20	STUDENT	Student / User department	192.168.20.0/24	192.168.20.1
30	GUEST	Guest / Public users	192.168.30.0/24	192.168.30.1
99	MGMT	Management (switch SVIs, admin host)	192.168.99.0/24	192.168.99.1
100	NATIVE	Native VLAN on all trunks (no hosts)	None (Layer 2 only)	None
999	BLACKHOLE	Dead-end VLAN for unused ports (all shut down)	None (Layer 2 only)	None

All user subnets use a /24 (255.255.255.0) mask, giving 254 usable addresses per VLAN, and the third octet of each subnet matches its VLAN ID so that addresses are self-documenting and easy to troubleshoot. VLAN 100 and VLAN 999 are intentionally pure Layer 2 constructs with no subnet, no gateway, and no hosts: VLAN 100 exists only so that untagged trunk traffic lands in an isolated VLAN, and VLAN 999 exists only to quarantine unused ports. In this context "unused" means a port that lies outside the departmental provisioning plan of Section 5.3; a provisioned department port that simply has no workstation attached at present remains in its own user VLAN so that the network can grow without reconfiguration.

5.2 IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0.10	192.168.10.1	255.255.255.0	N/A
R1	G0/0/0.20	192.168.20.1	255.255.255.0	N/A
R1	G0/0/0.30	192.168.30.1	255.255.255.0	N/A
R1	G0/0/0.99	192.168.99.1	255.255.255.0	N/A
R1	G0/0/1 (WAN)	209.165.200.225	255.255.255.252	N/A

ISP	G0/0/0 (WAN)	209.165.200.226	255.255.255.252	N/A
ISP	G0/0/1	209.165.201.1	255.255.255.224	N/A
SRV-WEB	NIC	209.165.201.10	255.255.255.224	209.165.201.1
S1	SVI VLAN 99	192.168.99.11	255.255.255.0	192.168.99.1
S2	SVI VLAN 99	192.168.99.12	255.255.255.0	192.168.99.1
ADMIN-PC	NIC (static)	192.168.99.21	255.255.255.0	192.168.99.1
PC-A / PC-D	NIC (DHCP)	From VLAN 10 pool	255.255.255.0	192.168.10.1
PC-B / PC-E	NIC (DHCP)	From VLAN 20 pool	255.255.255.0	192.168.20.1
PC-C	NIC (DHCP)	From VLAN 30 pool	255.255.255.0	192.168.30.1

5.3 Switch Port Assignment Table

Switch	Ports	Mode	VLAN	Connected To
S1	Fa0/1 - Fa0/8	Access	10	PC-A on Fa0/1; rest reserved for Staff
S1	Fa0/9 - Fa0/16	Access	20	PC-B on Fa0/9; rest reserved for Students
S1	Fa0/17 - Fa0/20	Access	30	PC-C on Fa0/17; rest reserved for Guests
S1	Fa0/21 - Fa0/22, G0/2	Access (shut)	999	Unused, administratively down
S1	Fa0/23 - Fa0/24	Trunk (Po1)	Native 100	S2 Fa0/23 - Fa0/24 (LACP EtherChannel)
S1	G0/1	Trunk	Native 100	R1 G0/0/0 (Router-on-a-Stick)
S2	Fa0/1 - Fa0/8	Access	10	PC-D on Fa0/1; rest reserved for Staff

S2	Fa0/9 - Fa0/16	Access	20	PC-E on Fa0/9; rest reserved for Students
S2	Fa0/17 - Fa0/20	Access	30	Reserved for Guests
S2	Fa0/21	Access	99	ADMIN-PC (management workstation)
S2	Fa0/22, G0/1 - G0/2	Access (shut)	999	Unused, administratively down
S2	Fa0/23 - Fa0/24	Trunk (Po1)	Native 100	S1 Fa0/23 - Fa0/24 (LACP EtherChannel)

Provisioned ports versus unused ports: the access-port ranges above are deliberately configured across their full width rather than only on the ports that currently have a device attached. In a small-to-medium enterprise every desk position in a department is expected to be occupied in normal operation, so each department is allocated a contiguous block of ports: Fa0/1 to Fa0/8 for Staff, Fa0/9 to Fa0/16 for Student and Fa0/17 to Fa0/20 for Guest, which is provisioned in advance so that a new workstation can simply be plugged into its department block and receive the correct VLAN, address and gateway with no further configuration. These ports are therefore provisioned and enabled, not unused, and they are the spare capacity that makes the design scalable; placing them in the blackhole VLAN 999 would defeat that purpose and would require a configuration change every time the organisation hires a member of staff. Only the ports that fall outside the departmental provisioning plan: Fa0/21 and Fa0/22 with G0/2 on S1, and Fa0/22 with G0/1 and G0/2 on S2, are genuinely unused, and those are the ports assigned to VLAN 999 and administratively shut down. It should also be noted that leaving a provisioned port empty does not leave it unprotected: sticky port security with violation shutdown is applied to every one of these ports, and PortFast with BPDU Guard err-disables any port on which a rogue switch attempts to join the spanning tree, so an unauthorised device connected to a vacant department port is denied access and the port is disabled. Finally, only one end device per VLAN was connected during the demonstration because a single host is sufficient to prove that VLAN, its DHCP scope and its routing path; populating every provisioned port with additional identical workstations would add no further test value and would only make the topology diagram harder to read.

5.4 DHCP Pool Design

Pool Name	Network	Excluded Addresses	Default Gateway	DNS Server
VLAN10-STAFF	192.168.10.0/24	192.168.10.1 - 192.168.10.10	192.168.10.1	209.165.201.10
VLAN20-STUDENT	192.168.20.0/24	192.168.20.1 - 192.168.20.10	192.168.20.1	209.165.201.10

VLAN30-GUEST	192.168.30.0/24	192.168.30.1 - 192.168.30.10	192.168.30.1	209.165.201.10

R1 acts as the centralized DHCP server for the three user VLANs, so clients receive their address, mask, gateway, and DNS server automatically as soon as they connect. The first ten addresses of every user subnet are excluded from the pools; this protects the gateway address and reserves room for printers, access points, or other statically addressed devices each department may add later. The Management VLAN 99 deliberately has no DHCP pool: every management address is assigned statically so that infrastructure addressing is predictable and an attacker cannot simply lease a management address by plugging into the network. A DHCP pool for VLAN 99 was technically possible and was considered, but it was rejected for two reasons. Management addressing must be predictable, because the switch SVIs at 192.168.99.11 and 192.168.99.12 are reached by fixed address for SSH administration and any monitoring or filtering would reference those addresses; and a management pool would automatically issue a valid, working management-network address to any device that reached VLAN 99, whereas with no pool an intruder must additionally discover the subnet, mask and gateway before it can communicate at all. At enterprise scale the mature alternative would be DHCP with MAC-to-address reservations, which keeps addressing centrally managed while remaining deterministic, but this is outside the scope of the SRWE curriculum and unnecessary for a single administrative workstation.

5.5 Routing Plan

Router	Route Type	Route Statement	Purpose
R1	Default route	ip route 0.0.0.0 0.0.0.0 209.165.200.226	Sends all traffic for unknown destinations (the internet) to the ISP.
ISP	Static return route	ip route 192.168.0.0 255.255.0.0 209.165.200.225	One summary route that returns traffic to all internal 192.168.x.0/24 subnets via R1.

Internally, R1 needs no static routes because every VLAN subnet is directly connected through its ROAS subinterface, and inter-VLAN traffic is routed between those connected networks. Externally, R1 carries a single default route pointing at the ISP, which is the correct design for a stub network with exactly one exit path. On the ISP router, one summary return route covering 192.168.0.0/16 is used instead of four individual /24 routes; this keeps the ISP routing table minimal and automatically covers any additional internal VLANs added in the future. Both static routes specify a next-hop IP address rather than an exit interface, which is best practice on multi-access Ethernet links because it avoids relying on Proxy ARP.

Inter-VLAN communication policy: in this phase, all four VLANs are permitted to communicate with each other and with the internet through R1, which satisfies the assignment requirement for inter-VLAN connectivity and keeps testing simple. We note as a documented future enhancement that a production deployment would apply an ACL on the Guest subinterface to restrict VLAN 30 to internet-only access, since traffic filtering with ACLs is outside the scope of this SRWE assignment.

5.6 Security Plan

To fully support the organization's upgrade and secure network, this design deliberately implements a comprehensive suite of eight access-layer security features. These are summarized below, together with the command used to verify each one.

#	Security Feature	Where Applied	Verification
1	Sticky port security (max 2 MACs, violation shutdown)	All in-use access ports on S1 and S2	show port-security interface fa0/1
2	Unused ports shut down	All unused ports on S1 and S2	show interfaces status
3	Unused ports assigned to dead-end VLAN 999	All unused ports on S1 and S2	show vlan brief
4	DTP disabled (switchport nonegotiate + static modes)	Every access and trunk port	show interfaces fa0/1 switchport
5	Native VLAN changed from VLAN 1 to VLAN 100	All trunk links	show interfaces trunk
6	PortFast with BPDU Guard	All access ports	show spanning-tree interface fa0/1 detail
7	Secure management VLAN 99 (separate from user VLANs)	S1 and S2 SVIs, ADMIN-PC	show ip interface brief
8	SSH v2 only, local login, Telnet disabled	VTY lines of R1, S1, S2 (Packet Tracer); routers only in the physical lab	show ip ssh / ssh from ADMIN-PC

Together these controls address the most common Layer 2 attacks:

- port security blocks CAM-table flooding and casual device swapping
- the dead-end VLAN and shutdown state stop anyone using a spare wall port
- disabling DTP and moving the native VLAN defeat switch-spoofing and double-tagging VLAN-hopping attacks
- BPDU Guard err-disables any port where a rogue switch attempts to join the spanning tree.

Note on the physical laboratory: SSH version 2 with local authentication is the most suitable and secure approach for remote device management, and it is implemented and verified on all four devices in the Packet Tracer build. In the physical laboratory the available Catalyst 2960 switches run an older IOS image that does not include the cryptographic feature set, so the crypto key generate rsa command is unavailable and SSH cannot be enabled on S1 and S2. On that equipment control 8 was therefore applied to the routers only, and the switch management plane was protected instead by

console and VTY line passwords, an encrypted enable secret, local user accounts, service password-encryption and the isolated Management VLAN 99. This limitation and the compensating controls are documented in Section 8.

6. Configuration Details

This section documents the exact IOS commands used for the Packet Tracer build (Stage 2). Commands are shown once per feature; where S2 mirrors S1, only the differing lines are listed. The complete running configurations captured after implementation will be attached in the Appendix.

6.1 Basic Device Hardening and SSH (all devices)

Shown for S1; repeat on S2, R1, and ISP with the matching hostname

```
Switch(config)# hostname S1
S1(config)# enable secret class
S1(config)# service password-encryption
S1(config)# banner motd #Authorized access only!#
S1(config)# username admin secret admin
S1(config)# ip domain-name sme.local
S1(config)# crypto key generate rsa general-keys modulus 2048
S1(config)# ip ssh version 2
S1(config)# line console 0
S1(config-line)# password cisco
S1(config-line)# login
S1(config-line)# exec-timeout 5 0
S1(config)# line vty 0 15
S1(config-line)# password cisco
S1(config-line)# login local
S1(config-line)# transport input ssh
```

Switch management addressing (SVI in VLAN 99):

```
S1(config)# interface vlan 99
S1(config-if)# ip address 192.168.99.11 255.255.255.0
S1(config-if)# no shutdown
S1(config)# ip default-gateway 192.168.99.1
! On S2 use: ip address 192.168.99.12 255.255.255.0
```

6.2 VLAN Creation and Access Ports

```
S1(config)# vlan 10
S1(config-vlan)# name STAFF
S1(config-vlan)# vlan 20
S1(config-vlan)# name STUDENT
S1(config-vlan)# vlan 30
S1(config-vlan)# name GUEST
S1(config-vlan)# vlan 99
S1(config-vlan)# name MGMT
S1(config-vlan)# vlan 100
S1(config-vlan)# name NATIVE
```

```
S1(config-vlan)# vlan 999
S1(config-vlan)# name BLACKHOLE
S1(config)# interface range fa0/1 - 8
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 10
S1(config-if-range)# switchport nonegotiate
S1(config)# interface range fa0/9 - 16
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config-if-range)# switchport nonegotiate
S1(config)# interface range fa0/17 - 20
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 30
S1(config-if-range)# switchport nonegotiate
! S2 is identical, plus the management access port:
S2(config)# interface fa0/21
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 99
S2(config-if-range)# switchport nonegotiate
```

6.3 Trunking and Native VLAN

Uplink from S1 to R1 (Router-on-a-Stick trunk):

```
S1(config)# interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 100
S1(config-if)# switchport trunk allowed vlan 10,20,30,99
S1(config-if)# switchport nonegotiate
```

Setting the mode statically to trunk and adding switchport nonegotiate disables DTP frames entirely, the allowed-VLAN list prunes everything except the four production VLANs, and the native VLAN is the unused VLAN 100 so untagged frames can never reach a user VLAN.

6.4 LACP EtherChannel and Spanning Tree

```
S1(config)# interface range fa0/23 - 24
S1(config-if-range)# channel-group 1 mode active
S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 100
S1(config-if)# switchport trunk allowed vlan 10,20,30,99
S1(config-if)# switchport nonegotiate
! Repeat identically on S2 (channel-group 1 mode active on fa0/23 - 24)
S1(config)# spanning-tree vlan 1,10,20,30,99,100,999 root primary
S2(config)# spanning-tree vlan 1,10,20,30,99,100,999 root secondary
S1(config)# interface range fa0/1 - 20
S1(config-if-range)# spanning-tree portfast
S1(config-if-range)# spanning-tree bpduguard enable
! On S2 apply portfast/bpduguard to fa0/1 - 21
```

Both sides use LACP active mode so the bundle always negotiates, and the trunk settings are applied on the port-channel interface so they propagate consistently to both member links. S1 is manually forced to be the Root Bridge for every VLAN, with S2 as the deterministic backup, so the topology never depends on default MAC-address-based root election. Spanning-tree mode: no spanning-tree

mode command appears above because PVST+ is the default spanning-tree mode on the Catalyst 2960, and it was deliberately retained rather than changed. The mode in use is confirmed by the show spanning-tree output in Figure 7.13, which reports "Spanning tree enabled protocol ieee" for every VLAN; ieee is the identifier Cisco IOS uses for the 802.1D per-VLAN instance, whereas Rapid PVST+ would instead report protocol rstp. A per-VLAN mode is a requirement of this design rather than an incidental default, because the root bridge is placed explicitly for each VLAN by the root primary and root secondary commands above; a single common spanning-tree instance such as classic 802.1D would make that per-VLAN placement impossible. Rapid PVST+ was considered as an alternative, since 802.1w converges in a few seconds against the thirty to fifty seconds taken by 802.1D. It was not adopted for two reasons. First, the only redundant path in this topology is the inter-switch link, and that redundancy is already provided by the LACP EtherChannel rather than by spanning tree: because Port-channel 1 is a single logical link, the loss of one member port redistributes traffic across the survivor without any spanning-tree recalculation at all, so the slower convergence of 802.1D is never exercised in normal operation. Second, every access port carries PortFast, so host ports transition to forwarding immediately and end users never experience the 802.1D listening and learning delay. Migrating to Rapid PVST+ would nonetheless be the correct next step if additional redundant switch-to-switch paths were introduced, and both switches would have to be changed together, since a mode mismatch causes the Rapid PVST+ switch to fall back to 802.1D-format BPDUs on the affected link and forfeits the benefit.

6.5 Inter-VLAN Routing (Router-on-a-Stick on R1)

```
R1(config)# interface g0/0/0
R1(config-if)# no shutdown
R1(config)# interface g0/0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config)# interface g0/0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config)# interface g0/0/0.30
R1(config-subif)# encapsulation dot1Q 30
R1(config-subif)# ip address 192.168.30.1 255.255.255.0
R1(config)# interface g0/0/0.99
R1(config-subif)# encapsulation dot1Q 99
R1(config-subif)# ip address 192.168.99.1 255.255.255.0
```

Each subinterface number matches its VLAN ID for readability. No subinterface is created for VLAN 100 or VLAN 999, so the native and blackhole VLANs have no Layer 3 gateway and can never be routed.

6.6 DHCP Configuration (on R1)

```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)# ip dhcp excluded-address 192.168.20.1 192.168.20.10
R1(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.10
R1(config)# ip dhcp pool VLAN10-STAFF
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
```

```
R1(dhcp-config)# dns-server 209.165.201.10
R1(dhcp-config)# domain-name sme.local
R1(config)# ip dhcp pool VLAN20-STUDENT
R1(dhcp-config)# network 192.168.20.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.20.1
R1(dhcp-config)# dns-server 209.165.201.10
R1(dhcp-config)# domain-name sme.local
R1(config)# ip dhcp pool VLAN30-GUEST
R1(dhcp-config)# network 192.168.30.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.30.1
R1(dhcp-config)# dns-server 209.165.201.10
R1(dhcp-config)# domain-name sme.local
```

6.7 WAN Addressing and Static/Default Routing

```
R1(config)# interface g0/0/1
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# no shutdown
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.200.226
ISP(config)# interface g0/0/0
ISP(config-if)# ip address 209.165.200.226 255.255.255.252
ISP(config-if)# no shutdown
ISP(config)# interface g0/0/1
ISP(config-if)# ip address 209.165.201.1 255.255.255.224
ISP(config-if)# no shutdown
ISP(config)# ip route 192.168.0.0 255.255.0.0 209.165.200.225
! Physical laboratory only - substitutes for SRV-WEB, which is not available in
the lab
ISP(config)# interface loopback 0
ISP(config-if)# ip address 209.165.201.10 255.255.255.224
```

6.8 Port Security and Unused Ports

```
S1(config)# interface range fa0/1 - 20
S1(config-if-range)# switchport port-security
S1(config-if-range)# switchport port-security maximum 2
S1(config-if-range)# switchport port-security mac-address sticky
S1(config-if-range)# switchport port-security violation shutdown
! On S2 apply the same to fa0/1 - 21
S1(config)# interface range fa0/21 - 22, g0/2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# switchport nonegotiate
S1(config-if-range)# shutdown
! On S2: interface range fa0/22, g0/1 - 2 (same commands)
```

6.9 End Device Configuration

The five user PCs require no typed network settings: each network adapter is simply set to obtain its address automatically (DHCP), and R1 supplies the IP address, subnet mask, default gateway, and DNS server for the PC's VLAN. This is deliberate as it demonstrates the centralized DHCP requirement and means a device moved to a different department port reconfigures itself with no administrator involvement. Only two endpoints are configured manually, as summarized below.

Endpoint	Setting	Value
PC-A, PC-B, PC-C, PC-D, PC-E	IP configuration mode	DHCP (automatic) - no other settings entered
ADMIN-PC	IP address / mask	192.168.99.21 / 255.255.255.0 (static)
ADMIN-PC	Default gateway / DNS	192.168.99.1 / 209.165.201.10
SRV-WEB	IP address / mask	209.165.201.10 / 255.255.255.224 (static)
SRV-WEB	Default gateway	209.165.201.1
SRV-WEB	Services	HTTP enabled; DNS enabled with A record www.sme.local = 209.165.201.10

ADMIN-PC is static because the Management VLAN intentionally has no DHCP pool (Section 5.4): infrastructure addresses must be predictable, and no unauthorized device should be able to lease its way onto the management network. SRV-WEB is static because it is a server that clients and DHCP option values refer to by a fixed address; its DNS and HTTP services also give the network a realistic name-resolution and web-browsing target for end-to-end testing.

7. Testing and Verification

The following test matrix was executed first in Packet Tracer (Stage 2) and then repeated on the physical equipment (Stage 3). The corresponding screenshots and command outputs for each test are presented in Appendix B, and any failures encountered are documented in Section 8 (Troubleshooting) together with the fix applied.

#	Test	Method / Command	Expected Result	Evidence
T1	DHCP address assignment	On PC-A to PC-E: ipconfig /renew; on R1: show ip dhcp binding	Each PC receives an address from its own VLAN pool with correct gateway and DNS; bindings visible on R1.	Fig 7.4 - 7.8
T2	Excluded addresses respected	show ip dhcp pool; inspect leased addresses	No client ever receives .1 to .10 of any subnet.	Fig 7.8
T3	Intra-VLAN connectivity across switches	PC-A ping PC-D (both VLAN 10)	Ping succeeds through the EtherChannel trunk without routing.	Fig 7.1
T4	Inter-VLAN routing	PC-A ping PC-B; PC-B ping PC-C	Pings succeed via R1 subinterfaces (ROAS).	Fig 7.2
T5	Internet / default route	PC-B ping 209.165.201.10; tracert 209.165.201.10	Ping succeeds; trace shows hops 192.168.20.1 then 209.165.200.226.	Fig 7.3
T6	Return route on ISP	ISP: show ip route	Static 192.168.0.0/16 via 209.165.200.225 present; R1 shows S* default route.	Fig 7.11.1 - 7.11.2
T7	VLAN database and port membership	S1/S2: show vlan brief	VLANs 10, 20, 30, 99, 100, 999 exist with correct names and port assignments.	Fig 7.9.1 - 7.9.2
T8	Trunk operation and native VLAN	S1/S2: show interfaces trunk	G0/1 and Po1 trunking; native VLAN 100;	Fig 7.10

			allowed list 10,20,30,99.	
T9	EtherChannel health	show etherchannel summary	Po1 flagged SU with Fa0/23 and Fa0/24 bundled (P) using LACP.	Fig 7.12
T10	STP root placement	show spanning-tree	S1 is root for all VLANs; S2 shows root ID pointing to S1.	Fig 7.13
T11	Redundancy failover	Disconnect Fa0/23 during a continuous ping PC-A to PC-D	Ping continues over the remaining member link with no STP reconvergence.	Fig 7.14
T12	Port security	show port-security interface fa0/1; then swap PC-A for another device	Sticky MAC recorded; port goes err-disabled on violation.	Fig 7.15.1 - 7.15.4
T13	BPDU Guard	Connect a test switch to an access port	Port immediately err-disables on receiving a BPDU.	Fig 7.18.1 - 7.18.2
T14	Secure management	From ADMIN-PC: ssh -l admin 192.168.99.11; then attempt telnet	SSH v2 login succeeds; Telnet connection is refused.	Fig 7.16
T15	Unused port quarantine	show interfaces status; plug a PC into Fa0/22 on S1	Port shows disabled in VLAN 999; the device gets no connectivity.	Fig 7.17

8. Troubleshooting

This section records the genuine problems encountered during the Packet Tracer and real lab implementations, how each problem was identified using IOS verification commands, and how it was resolved.

#	Problem / Symptom	How We Identified It	How We Solved It
1.	ADMIN-PC (VLAN 99 management workstation) failed to obtain a working IP address. Its adapter was set to DHCP but it self-assigned a 169.254.x.x address and could not reach either switch SVI, so SSH management testing was blocked.	show vlan brief on S2 confirmed Fa0/21 was correctly assigned to VLAN 99, and show interfaces status showed the port connected, ruling out a Layer 1 or VLAN membership fault. Reviewing our own DHCP design in show ip dhcp pool on R1 confirmed the cause: only VLAN 10, 20 and 30 have DHCP pools. VLAN 99 was deliberately excluded from DHCP so that management addressing stays predictable, so no server existed to answer the request.	Reconfigured ADMIN-PC from DHCP to static addressing: 192.168.99.21, mask 255.255.255.0, gateway 192.168.99.1, DNS 209.165.201.10. Verified with ping 192.168.99.11 and ping 192.168.99.12 to both switch SVIs, which then succeeded. This matched the original design in Section 5.2, where ADMIN-PC is specified as statically addressed.
2.	SSH could not be configured on the physical lab switches. The crypto key generate rsa command was rejected, so no RSA key pair could be created and remote encrypted management could not be enabled, despite the same configuration working in Packet Tracer.	Confirmed hostname and ip domain-name sme.local were already set, eliminating the usual prerequisite error. show version revealed the switches were running an older IOS image without the cryptographic feature set, and show ip ssh returned no SSH support. This identified a hardware and image limitation rather than a configuration mistake.	Documented the limitation and applied the strongest available alternative on the physical switches: console line password protection, enable secret, local user accounts, service password-encryption, and a dedicated management VLAN 99 restricted to the ADMIN-PC. Full SSH v2 configuration and verification were retained and demonstrated in the Cisco Packet Tracer implementation, where the simulated IOS supports the crypto feature set.
3.	Several access ports that had been configured with port security did not appear in the port-security output at all. Only the ports with a device actively sending	show running-config interface fa0/21 confirmed that all four port-security commands were present on the interface, and show interfaces status reported	Traffic was generated from each attached host by pinging its default gateway, after which the sticky MAC address was learned and every port appeared in the output

#	Problem / Symptom	How We Identified It	How We Solved It
	traffic were listed, which raised doubt over whether the access-layer security configuration had been applied to the whole range or had silently failed on the remaining ports.	the port as connected, so neither the configuration nor the physical link was at fault. Comparing the port against Fa0/1, which did appear in the output, isolated the difference: Fa0/1 had already forwarded user traffic and had therefore learned a sticky MAC address, while the ports that were missing had not yet carried a single frame. We also confirmed that show port-security and show port-security address are two different commands, the first listing the ports on which the feature is enabled and the second listing only the secure addresses actually learned.	with its secure address. This confirmed that the configuration had been correct throughout. The group recorded the result as a verification lesson rather than a configuration fault: sticky learning is triggered by traffic, not by the command, so a port must forward at least one frame before its secure address can be displayed. Verification commands must therefore be read as a report of what a device has learned, which is not always the same as what it has been told to do.

9. Video Demonstration

The demonstration video is available at: https://youtu.be/uDCfK2LEvJc

10. Conclusion

This project successfully replaced an inefficient, flat single-broadcast-domain network with a segmented, secure and redundant small-to-medium enterprise design, implemented first in Cisco Packet Tracer and then on physical Cisco equipment in the laboratory. The final network delivers four departmental VLANs, Staff, Student, Guest and Management, plus a dedicated unused native VLAN and a blackhole VLAN for disabled ports, with inter-VLAN communication provided by 802.1Q Router-on-a-Stick on the core router. Centralised DHCP pools on R1 automatically address every user VLAN while reserving infrastructure addresses through excluded ranges, and a static default route paired with a summarised return route on the simulated ISP router establishes verified external connectivity. Redundancy between the switches is provided by a two-member LACP EtherChannel operating as a single logical link, with PVST+ root placement assigned deliberately rather than left to default election. Eight access-layer security controls were applied against the four required, sticky port security with violation shutdown, unused ports shut down and quarantined in a dead-end VLAN, DTP disabled, the native VLAN moved off VLAN 1, PortFast with BPDU Guard, an isolated management VLAN, and encrypted management access, and the planned verification tests were executed and evidenced with command output in both environments.

Working across simulation and real hardware taught the group that a design which passes in Packet Tracer does not transfer to physical equipment unchanged. Three constraints forced documented adaptations: the laboratory switches run an older IOS image without the cryptographic feature set, so SSH could not be enabled there and the management plane was instead protected by line passwords, an enable secret, local accounts and the isolated Management VLAN, with full SSH version 2 demonstrated in Packet Tracer; the absence of a physical server meant the external host was represented by a loopback interface on the ISP router, which answered ping and traceroute identically; and limited PC availability reduced the end-device count, so ports were allocated to keep exactly one device in every VLAN and preserve full test coverage. The group also learned that verification commands are diagnostic tools rather than a final formality. Several faults were only explicable through them: the management workstation failed to obtain an address until we recognised that VLAN 99 has no DHCP pool by design and that static addressing was always the intended configuration, and a port that appeared correctly configured was absent from the show port-security output because sticky learning registers a port only after it carries traffic. Adopting a bottom-up approach, confirming physical link state, then VLAN and trunk membership, then Layer 3 addressing, consistently located the true cause faster than inspecting configuration alone. Finally, dividing the work by stage while requiring every member to understand the complete design proved essential, because faults rarely stayed inside one person’s assigned area.

Two enhancements are documented as the natural next steps beyond the scope of this assignment. An extended access control list on the Guest subinterface would restrict VLAN 30 to internet-only access, and a second core router running an FHRP such as HSRP would remove the single point of failure that the core router and its trunk currently represent.

11. References

Cisco Networking Academy. (2020). *Switching, routing, and wireless essentials* (Version 7.0) [Online course]. Cisco. https://www.netacad.com

Cisco Systems. (n.d.). *Product support and documentation*. Cisco. Retrieved July 30, 2026, from https://www.cisco.com/c/en/us/support/index.html

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12. Appendix

Appendix A: Full Device Configurations

A.1 S1 Running Configuration

```
S1#show running-config
Building configuration...

Current configuration : 6932 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname S1
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name sme.local
!
username admin secret 5 $1$mERr$vTbHul1N28cEp8lkLqr0f/
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1,10,20,30,99-100,999 priority 24576
!
interface Port-channel1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0000.0C71.BA83
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/7
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/8
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0002.4A14.0792
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
!
interface FastEthernet0/10
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/11
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/12
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/14
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/15
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/16
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
!
interface FastEthernet0/17
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0007.EC1E.E6E7
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/18
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/19
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/20
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/21
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/22
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/23
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
 channel-group 1 mode active
!
interface FastEthernet0/24
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
 channel-group 1 mode active
```

```
!
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
!
interface GigabitEthernet0/2
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan99
 ip address 192.168.99.11 255.255.255.0
!
ip default-gateway 192.168.99.1
!
banner motd ^CAuthorized access only!^C
!
!
!
line con 0
 password 7 0822455D0A16
 login
 exec-timeout 5 0
!
line vty 0 4
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
!
!
!
!
end
```

A.2 S2 Running Configuration

```
S2#show running-config
Building configuration...

Current configuration : 7023 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname S2
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
ip ssh version 2
ip domain-name sme.local
!
username admin secret 5 $1$mERr$vTbHul1N28cEp8lkLqr0f/
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1,10,20,30,99-100,999 priority 28672
!
interface Port-channel1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0000.0C27.340C
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/3
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/7
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/8
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0001.971C.BB2C
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/10
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
```

```
!
interface FastEthernet0/11
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/12
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/13
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/14
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/15
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/16
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/17
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/18
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/19
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/20
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/21
 switchport access vlan 99
 switchport mode access
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0004.9AED.5A8C
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface FastEthernet0/22
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface FastEthernet0/23
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
 channel-group 1 mode active
!
interface FastEthernet0/24
 switchport trunk native vlan 100
 switchport trunk allowed vlan 10,20,30,99
 switchport mode trunk
 switchport nonegotiate
 channel-group 1 mode active
!
interface GigabitEthernet0/1
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
```

```
!
interface GigabitEthernet0/2
 switchport access vlan 999
 switchport mode access
 switchport nonegotiate
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan99
 ip address 192.168.99.12 255.255.255.0
!
ip default-gateway 192.168.99.1
!
banner motd ^CAuthorized access only!^C
!
!
!
line con 0
 password 7 0822455D0A16
 login
 exec-timeout 5 0
!
line vty 0 4
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
!
!
!
!
end
```

A.3 R1 Running Configuration

```
R1#show running-config
Building configuration...

Current configuration : 2033 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
!
ip dhcp pool VLAN10-STAFF
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 209.165.201.10
 domain-name sme.local
ip dhcp pool VLAN20-STUDENT
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 209.165.201.10
 domain-name sme.local
ip dhcp pool VLAN30-GUEST
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 209.165.201.10
 domain-name sme.local
!
!
!
no ip cef
no ipv6 cef
!
!
!
username admin secret 5 $1$mERr$vTbHul1N28cEp81kLqr0f/
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name sme.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
!
interface GigabitEthernet0/0/0.99
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
!
interface GigabitEthernet0/0/1
 ip address 209.165.200.225 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 209.165.200.226
!
ip flow-export version 9
!
!
!
banner motd ^CAuthorized access only!^C
!
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 login
!
!
```

A.4 ISP Running Configuration

```
ISP#show running-config
Building configuration...

Current configuration : 1092 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ISP
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username admin secret 5 $1$mERr$vTbHul1N28cEp8lkLqr0f/
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip domain-name sme.local
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0/0
 ip address 209.165.200.226 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 ip address 209.165.201.1 255.255.255.224
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
```

```
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.0.0 255.255.0.0 209.165.200.225
!
ip flow-export version 9
!
!
!
banner motd ^CAuthorized access only!^C
!
!
!
!
line con 0
 exec-timeout 5 0
 password 7 0822455D0A16
 login
!
line aux 0
!
line vty 0 4
 access-class MGMT-SSH in
 password 7 0822455D0A16
 login local
 transport input ssh
line vty 5 15
 access-class MGMT-SSH in
 login
!
!
!
end
```

Appendix B: Additional Screenshots

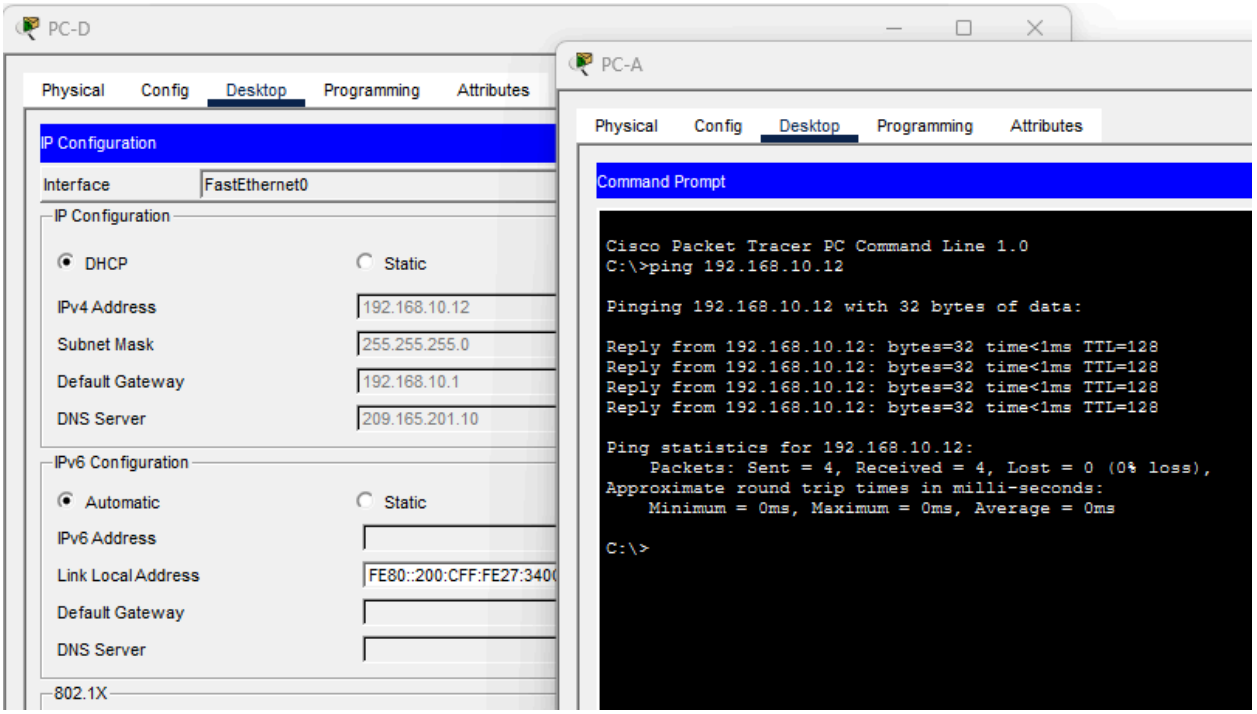


Figure 7.1: T3 (Packet Tracer) - PC-A pings PC-D, proving Intra-VLAN across Port-channel 1

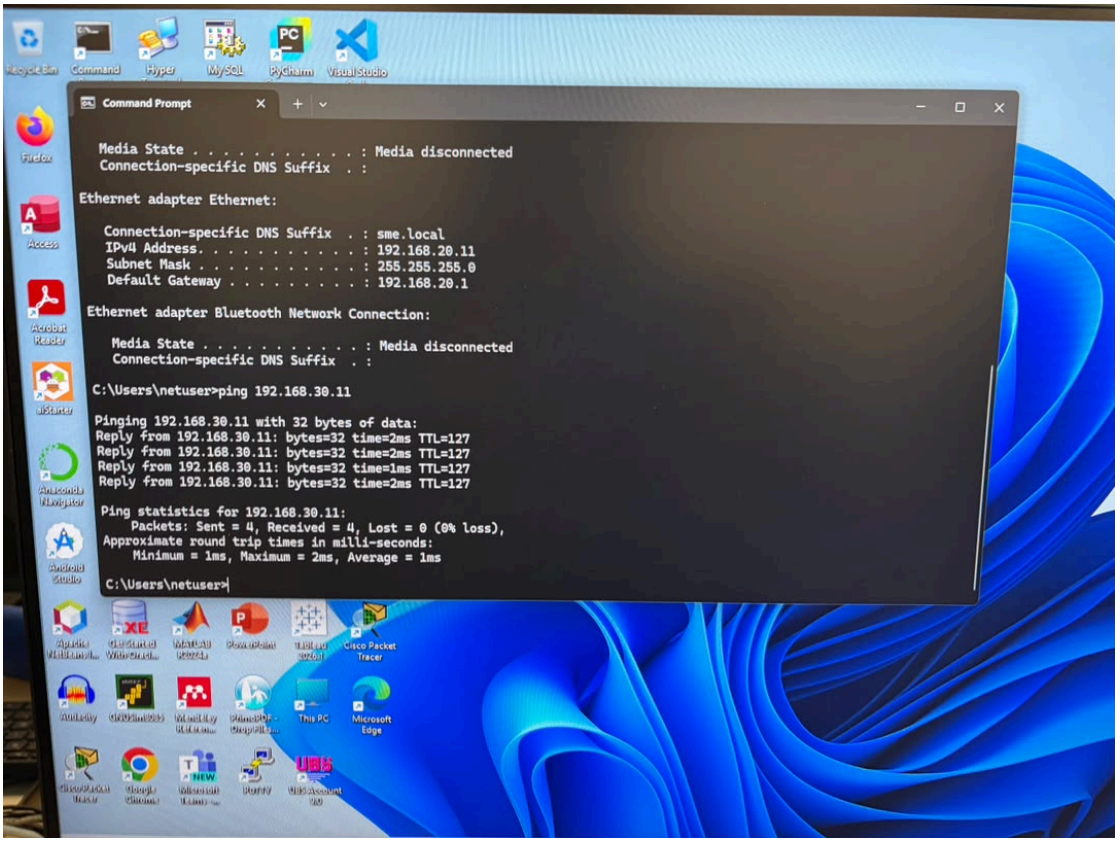


Figure 7.2: T4 - PC-B pings PC-C, Inter-VLAN working across EtherChannel

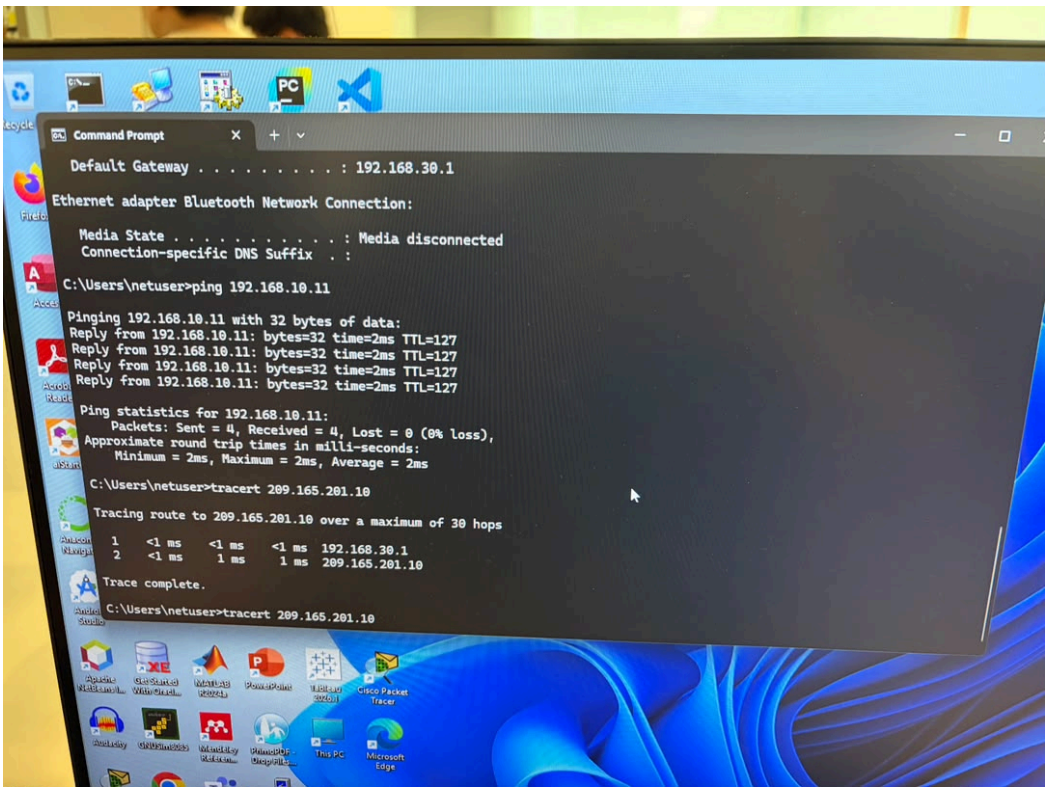


Figure 7.3: T5 - tracert 209.165.201.10, the default route and the return route

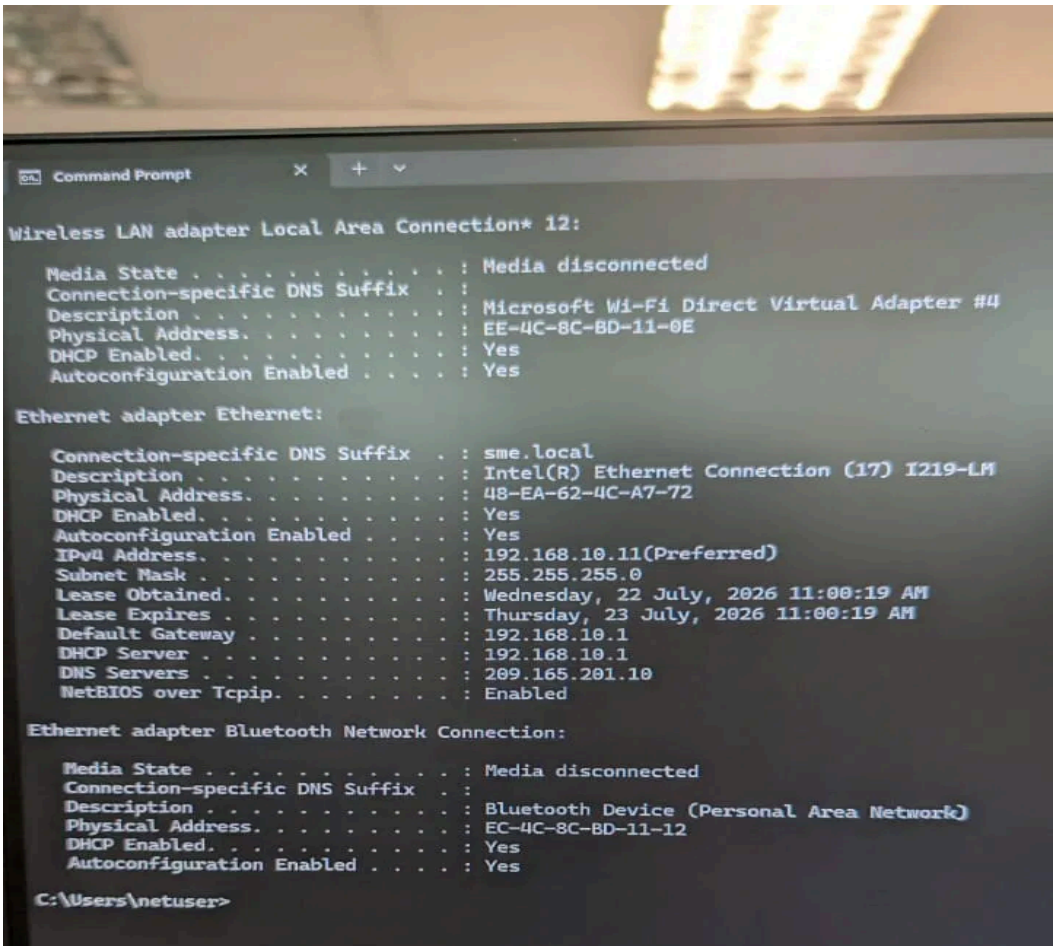


Figure 7.4: T1 - ipconfig on PC-A

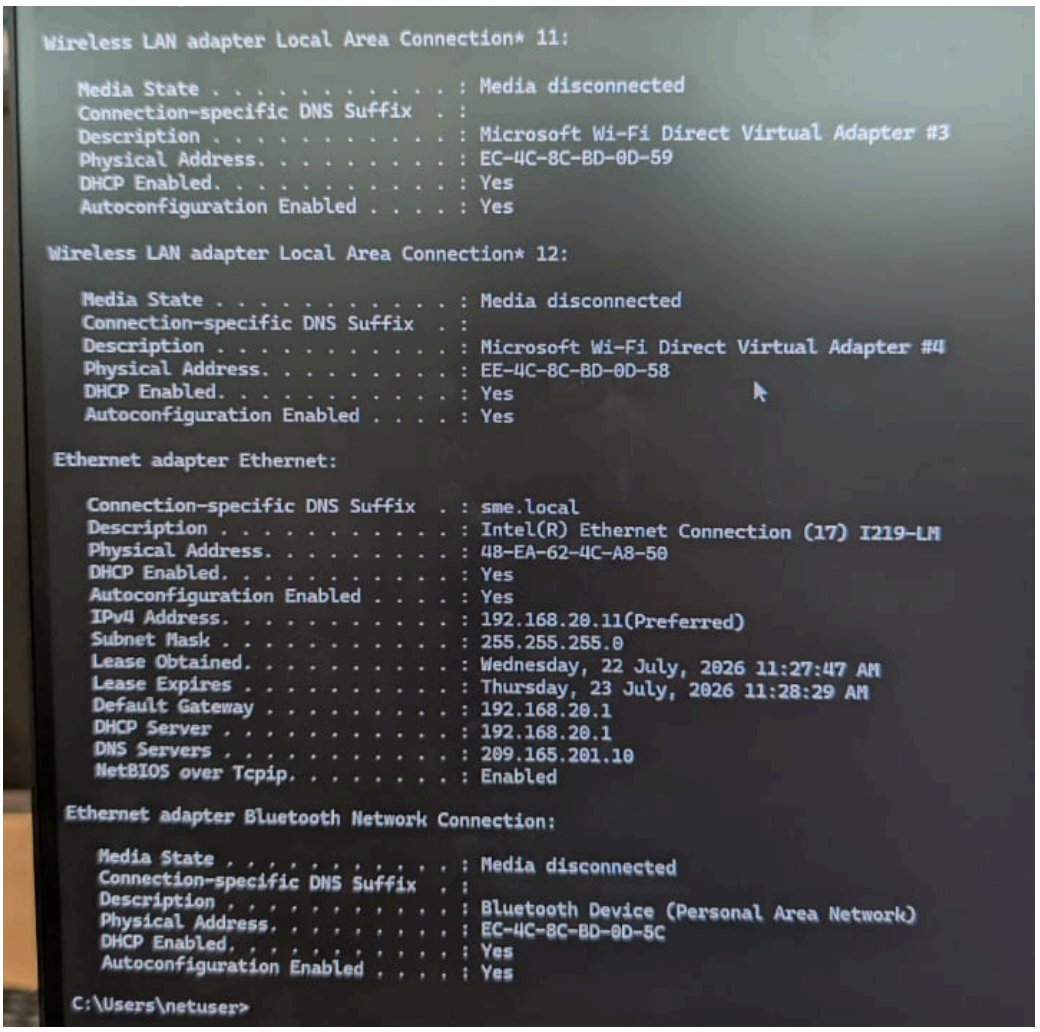


Figure 7.5: T1 - ipconfig on PC-B

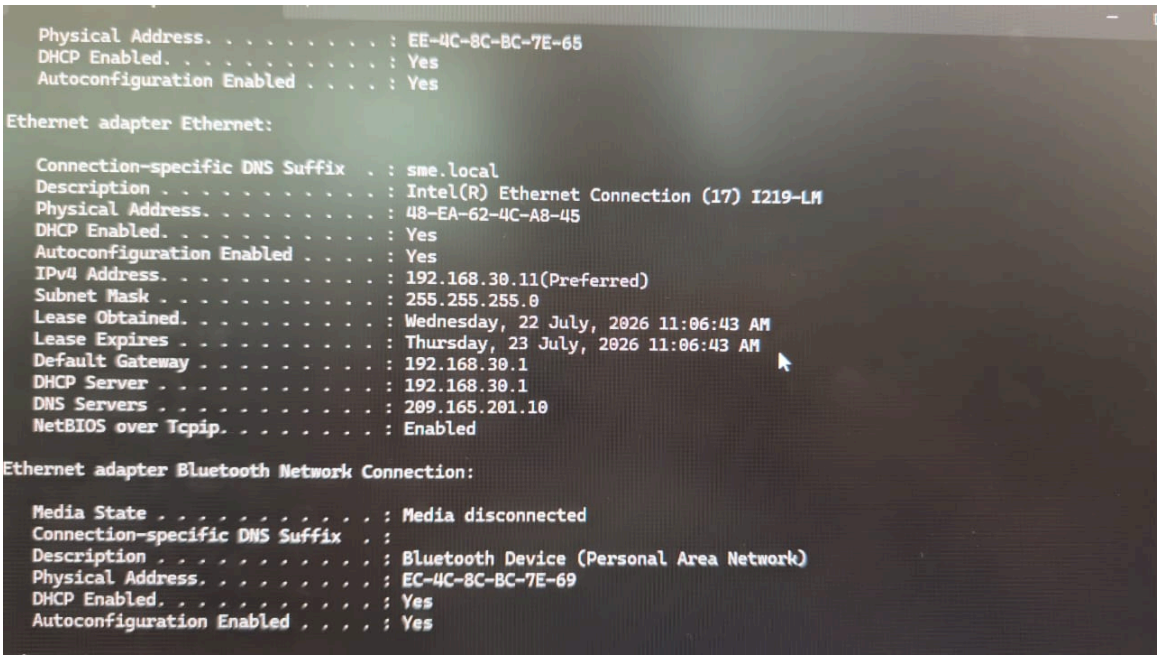


Figure 7.6: T1 - ipconfig on PC-C

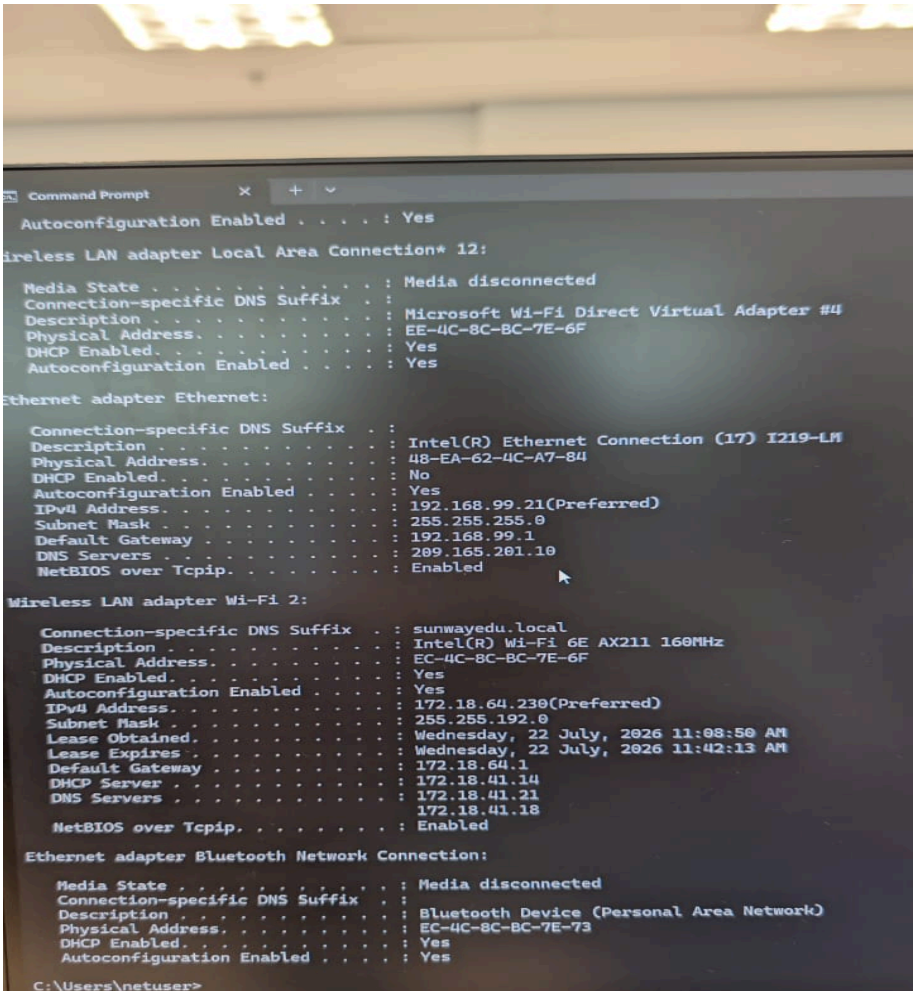


Figure 7.7: T1 - ipconfig on ADMIN-PC

```
R1#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.11    0000.0C71.BA83           --                     Automatic
192.168.10.12    0000.0C27.340C           --                     Automatic
192.168.20.11    0002.4A14.0792           --                     Automatic
192.168.20.12    0001.971C.BB2C           --                     Automatic
192.168.30.11    0007.EC1E.E6E7           --                     Automatic
R1#
```

```
R1#show ip dhcp pool

Pool VLAN10-STAFF :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    2    / 3     / 254

Pool VLAN20-STUDENT :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.20.1         192.168.20.1     - 192.168.20.254    2    / 3     / 254

Pool VLAN30-GUEST :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 3
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.30.1         192.168.30.1     - 192.168.30.254    1    / 3     / 254
R1#
```

Figure 7.8: T1/T2 - R1 show ip dhcp binding + show ip dhcp pool

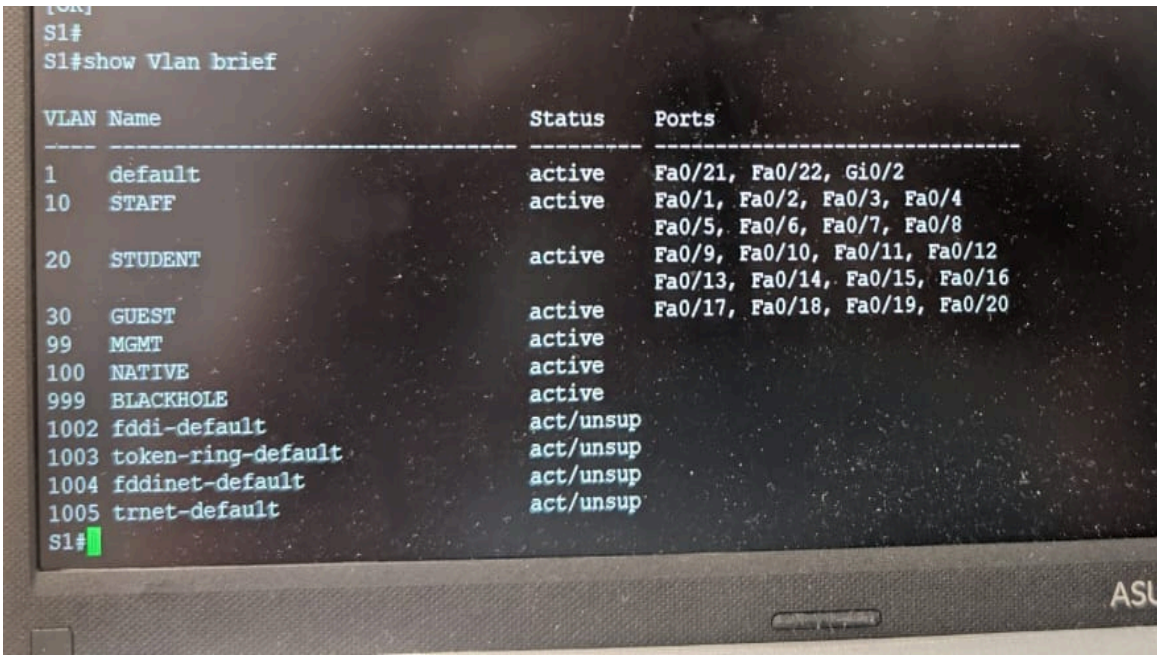


Figure 7.9.1: T7 - show vlan brief from S1

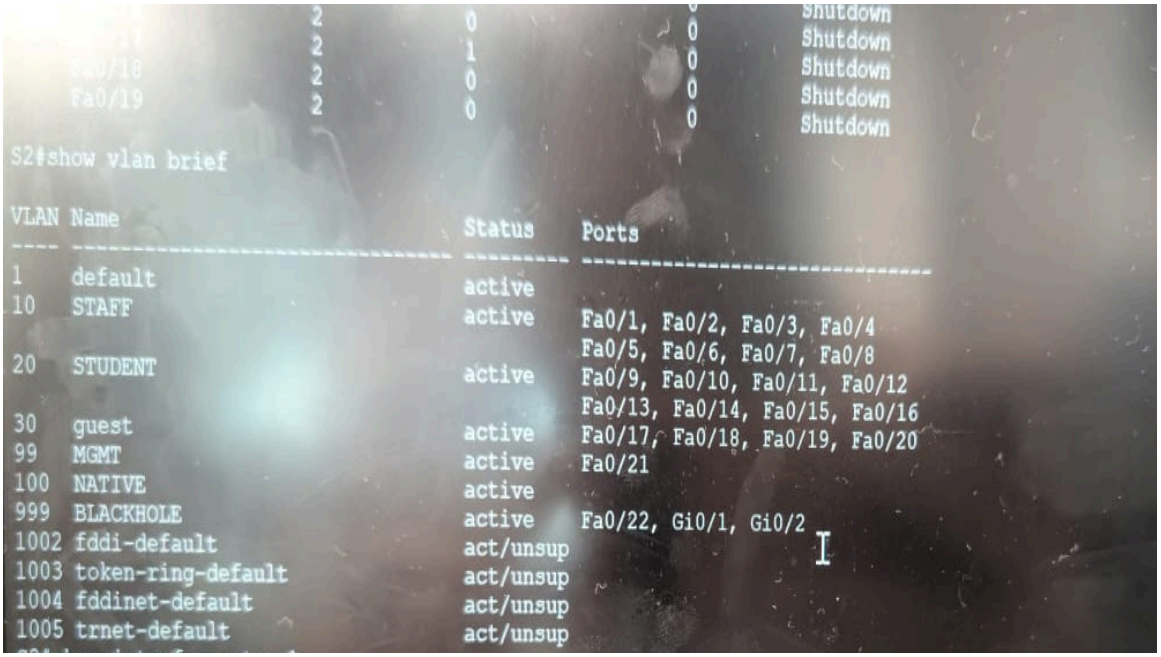


Figure 7.9.2: T7 - show vlan brief from S2

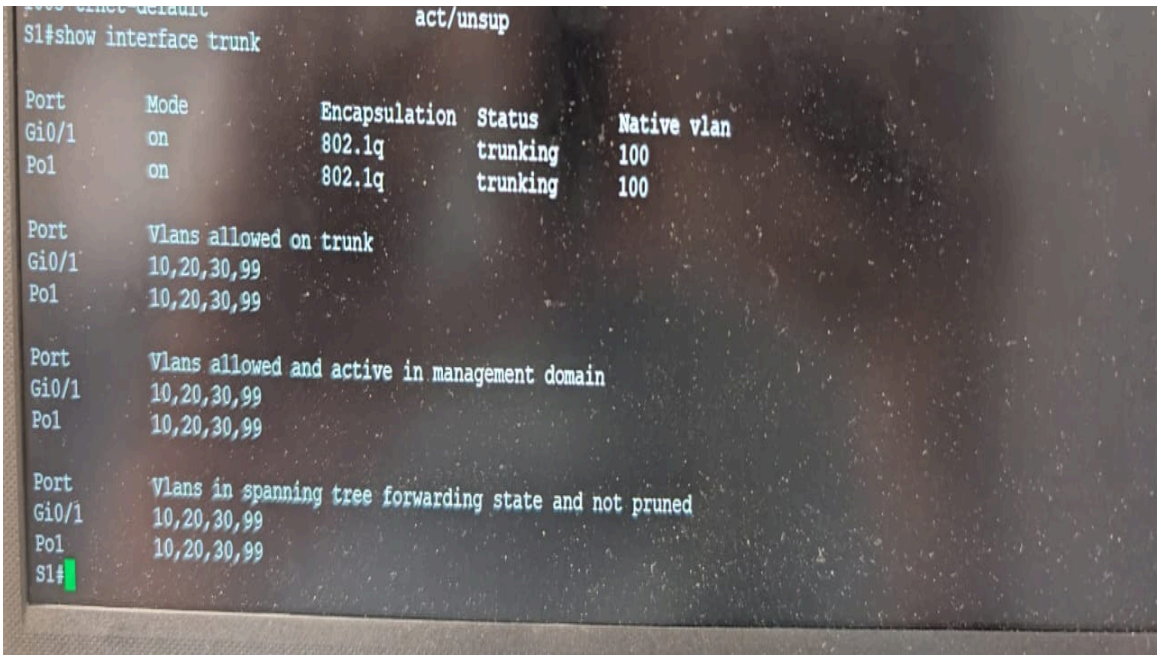


Figure 7.10: T8 - show interfaces trunk on S1

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 209.165.200.226 to network 0.0.0.0

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0/0.30
     192.168.99.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.99.0/24 is directly connected, GigabitEthernet0/0/0.99
L       192.168.99.1/32 is directly connected, GigabitEthernet0/0/0.99
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L       209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 [1/0] via 209.165.200.226
```

Figure 7.11.1: T6 - R1 show ip route

```
ISP#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.0.0/16 [1/0] via 209.165.200.225
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/0
L       209.165.200.226/32 is directly connected, GigabitEthernet0/0/0
     209.165.201.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.201.0/27 is directly connected, GigabitEthernet0/0/1
L       209.165.201.1/32 is directly connected, GigabitEthernet0/0/1

ISP#
```

Figure 7.11.2: T6 - ISP show ip route

```
S1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
S1#
```

Figure 7.12: T9 - S1 show etherchannel summary

```
S1#show spanning-tree
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0005.5EA8.E893
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0005.5EA8.E893
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Po1              Desg FWD 12        128.27   P2p

VLAN0020
  Spanning tree enabled protocol ieee
  Root ID    Priority    24596
             Address     0005.5EA8.E893
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24596  (priority 24576 sys-id-ext 20)
             Address     0005.5EA8.E893
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/9            Desg FWD 19        128.9    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Po1              Desg FWD 12        128.27   P2p

VLAN0030
  Spanning tree enabled protocol ieee
  Root ID    Priority    24606
             Address     0005.5EA8.E893
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24606  (priority 24576 sys-id-ext 30)
             Address     0005.5EA8.E893
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/17           Desg FWD 19        128.17   P2p
Gi0/1            Desg FWD 4         128.25   P2p
Po1              Desg FWD 12        128.27   P2p

VLAN0099
  Spanning tree enabled protocol ieee
  Root ID    Priority    24675
             Address     0005.5EA8.E893
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24675  (priority 24576 sys-id-ext 99)
             Address     0005.5EA8.E893
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Gi0/1            Desg FWD 4         128.25   P2p
Po1              Desg FWD 12        128.27   P2p

VLAN0100
  Spanning tree enabled protocol ieee
  Root ID    Priority    24676
             Address     0005.5EA8.E893
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24676  (priority 24576 sys-id-ext 100)
             Address     0005.5EA8.E893
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/23           Desg FWD 19        128.23   P2p
Gi0/1            Desg FWD 4         128.25   P2p
Fa0/24           Desg FWD 19        128.24   P2p
Po1              Desg FWD 12        128.27   P2p

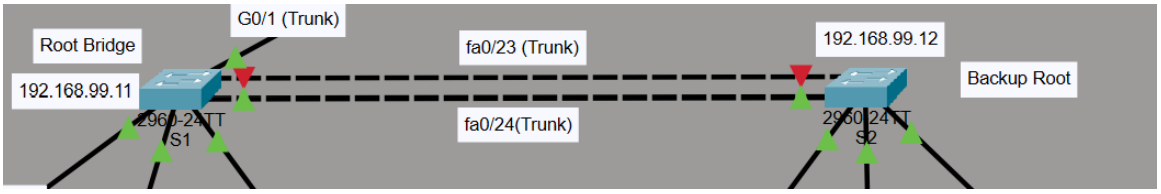
S1#
```

Figure 7.13: T10 - S1 show spanning- tree

```
C:\>ping -t 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
```




```
S1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#int f0/23
S1(config-if)#shut

S1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP    Fa0/23(D) Fa0/24(P)
```

Figure 7.14: T11 - (Packet Tracer)Failover pair and S1 show etherchannel summary for redundancy test

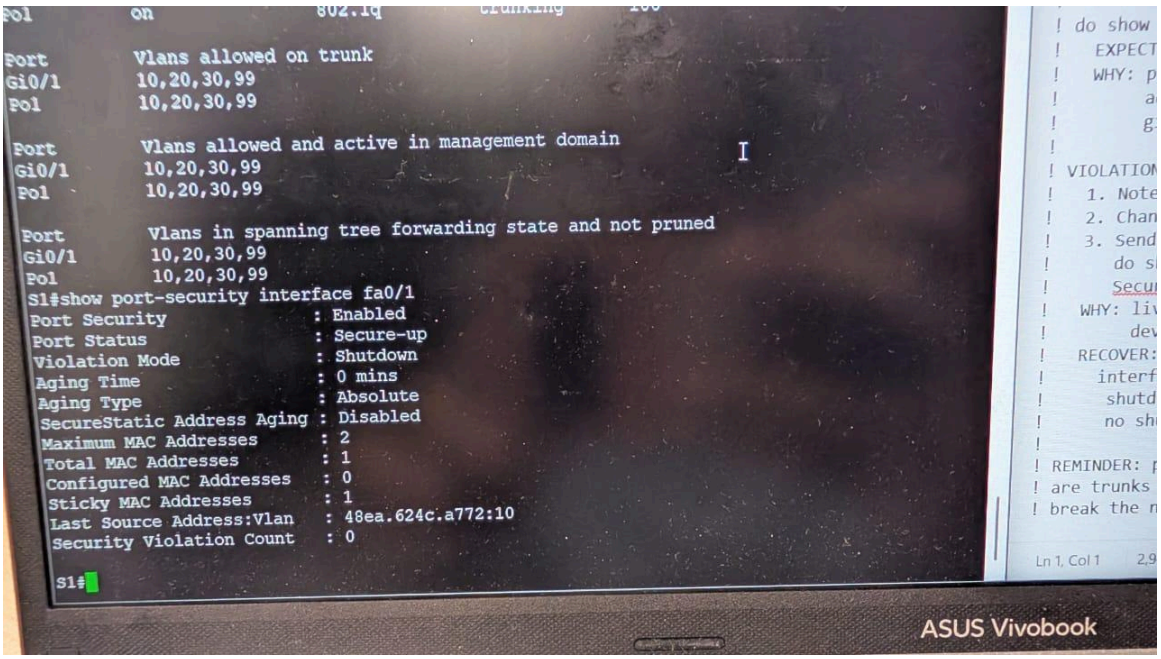


Figure 7.15.1: T12 - show port-security interface fa0/1 BEFORE violation

```
S1#show port-security int fa0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0000.0C71.BA83:10
Security Violation Count   : 0
```

Figure 7.15.2: T12 - (Packet Tracer) port-security int fa0/1 on PC-A

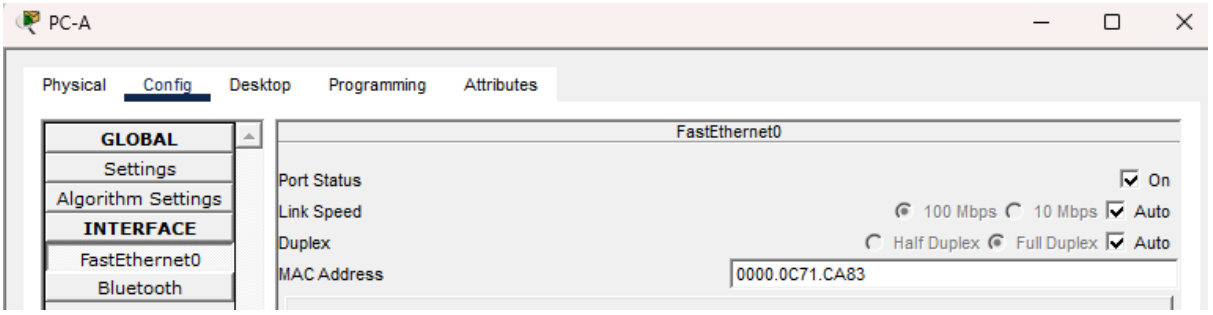


Figure 7.15.3: T12 - (Packet Tracer) changed the Test-PC's mac address for testing

```
S1#show port-security int fa0/1
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : 0000.0C71.CA83:10
Security Violation Count   : 1
```

Figure 7.15.4: T12 - (Packet Tracer) show port-security interface fa0/1 AFTER violation

```
C:\>ssh -l admin 192.168.99.11

Password:

Authorized access only!

S1>exit

[Connection to 192.168.99.11 closed by foreign host]
C:\>telnet 192.168.99.11
Trying 192.168.99.11 ...Open

[Connection to 192.168.99.11 closed by foreign host]
C:\>
```

Figure 7.16: T14 - SSH login + Telnet refused from ADMIN-PC

```
S1#show int status
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        notconnect   10         auto    auto  10/100BaseTX
Fa0/3                        notconnect   10         auto    auto  10/100BaseTX
Fa0/4                        notconnect   10         auto    auto  10/100BaseTX
Fa0/5                        notconnect   10         auto    auto  10/100BaseTX
Fa0/6                        notconnect   10         auto    auto  10/100BaseTX
Fa0/7                        notconnect   10         auto    auto  10/100BaseTX
Fa0/8                        notconnect   10         auto    auto  10/100BaseTX
Fa0/9                        connected    20         a-full  a-100 10/100BaseTX
Fa0/10                       notconnect   20         auto    auto  10/100BaseTX
Fa0/11                       notconnect   20         auto    auto  10/100BaseTX
Fa0/12                       notconnect   20         auto    auto  10/100BaseTX
Fa0/13                       notconnect   20         auto    auto  10/100BaseTX
Fa0/14                       notconnect   20         auto    auto  10/100BaseTX
Fa0/15                       notconnect   20         auto    auto  10/100BaseTX
Fa0/16                       notconnect   20         auto    auto  10/100BaseTX
Fa0/17                       connected    30         a-full  a-100 10/100BaseTX
Fa0/18                       notconnect   30         auto    auto  10/100BaseTX
Fa0/19                       notconnect   30         auto    auto  10/100BaseTX
Fa0/20                       notconnect   30         auto    auto  10/100BaseTX
Fa0/21                       disabled 999          auto    auto  10/100BaseTX
Fa0/22                       disabled 999          auto    auto  10/100BaseTX
Fa0/23                       connected    trunk      a-full  a-100 10/100BaseTX
Fa0/24                       connected    trunk      a-full  a-100 10/100BaseTX
Gig0/1                       connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig0/2                       disabled 999          auto    auto  10/100/1000BaseTX
```

```
S2#show int status
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        notconnect   10         auto    auto  10/100BaseTX
Fa0/3                        notconnect   10         auto    auto  10/100BaseTX
Fa0/4                        notconnect   10         auto    auto  10/100BaseTX
Fa0/5                        notconnect   10         auto    auto  10/100BaseTX
Fa0/6                        notconnect   10         auto    auto  10/100BaseTX
Fa0/7                        notconnect   10         auto    auto  10/100BaseTX
Fa0/8                        notconnect   10         auto    auto  10/100BaseTX
Fa0/9                        connected    20         a-full  a-100 10/100BaseTX
Fa0/10                       notconnect   20         auto    auto  10/100BaseTX
Fa0/11                       notconnect   20         auto    auto  10/100BaseTX
Fa0/12                       notconnect   20         auto    auto  10/100BaseTX
Fa0/13                       notconnect   20         auto    auto  10/100BaseTX
Fa0/14                       notconnect   20         auto    auto  10/100BaseTX
Fa0/15                       notconnect   20         auto    auto  10/100BaseTX
Fa0/16                       notconnect   20         auto    auto  10/100BaseTX
Fa0/17                       notconnect   30         auto    auto  10/100BaseTX
Fa0/18                       notconnect   30         auto    auto  10/100BaseTX
Fa0/19                       notconnect   30         auto    auto  10/100BaseTX
Fa0/20                       notconnect   30         auto    auto  10/100BaseTX
Fa0/21                       connected    99         a-full  a-100 10/100BaseTX
Fa0/22                       disabled 999          auto    auto  10/100BaseTX
Fa0/23                       connected    trunk      a-full  a-100 10/100BaseTX
Fa0/24                       connected    trunk      a-full  a-100 10/100BaseTX
Gig0/1                       disabled 999          auto    auto  10/100/1000BaseTX
Gig0/2                       disabled 999          auto    auto  10/100/1000BaseTX
```

Figure 7.17: T15 - show interfaces status on S1 + S2, showing unused ports "disabled" in VLAN 999

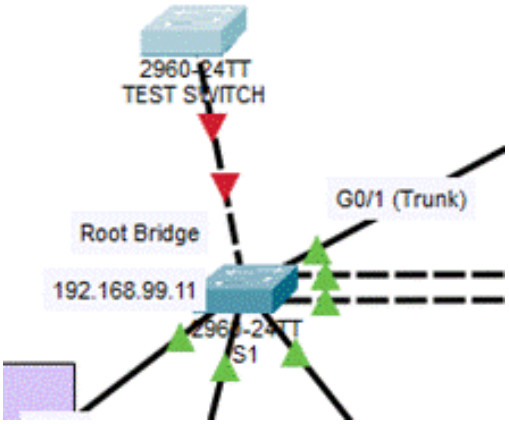


Figure 7.18.1: T13 (Packet Tracer) - TEST SWITCH connected to S1 Fa0/2 to generate BPDUs. The link is shown red because BPDU Guard has already err-disabled the port.

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (10), with Switch FastEthernet
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port FastEthernet0/2 with BPDU Guard enabled. Disabling port.

%PM-4-ERR_DISABLE: bpduguard error detected on 0/2, putting 0/2 in err-disable state


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

S1#show int
S1#show interfaces sta
S1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Po1                          connected    trunk      a-full  a-100
Fa0/1                        connected    10         a-full  a-100 10/100BaseTX
Fa0/2                        err-disabled 10         auto    auto  10/100BaseTX
```

Figure 7.18.2: T13 (Packet Tracer) - BPDU Guard operation on S1 Fa0/2. The console logs %SPANTREE-2-BLOCK_BPDUGUARD on receiving a BPDU and %PM-4-ERR_DISABLE places the port in err-disable state; show interfaces status then reports Fa0/2 as err-disabled in VLAN 10.

Figure 7.18.2 confirms that the err-disable reason is bpduguard rather than a port-security violation, which proves that BPDU Guard, and not port security, is the control that blocked the rogue switch. The CDP native VLAN mismatch message immediately above it is expected: the test switch was left at its factory default of native VLAN 1 while S1 Fa0/2 is an access port in VLAN 10. It is an artefact of the deliberately unconfigured rogue device used to generate the BPDU, not a fault in the production configuration, and it disappears with the port. Security control 6 in Section 5.6 is verified by this figure.

```
S1#show int fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (STAFF)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Figure 7.19: Security control 4 (Packet Tracer) - show interfaces fa0/1 switchport on S1, confirming Negotiation of Trunking: Off after switchport nonegotiate was applied.

Figure 7.19 is the verification listed for security control 4 in Section 5.6. Administrative Mode and Operational Mode both read static access, so the port can never negotiate itself into a trunk, and Negotiation of Trunking: Off confirms that the interface no longer transmits DTP frames at all. Together these close the switch-spoofing path used in VLAN hopping attacks. The Trunking Native Mode VLAN field still reports VLAN 1 because that value applies only when a port is trunking; on a static access port it is inert, and the native VLAN actually in use on every trunk is VLAN 100, as shown in Figure 7.10.